

ORIGINAL RESEARCH

CORONARY

Prognostic Value of Microvascular Resistance Reserve in Coronary Artery Disease



A Systematic Review and Meta-Analysis

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ABSTRACT

BACKGROUND Microvascular resistance reserve (MRR) is a novel index for evaluating coronary microvascular function independently of epicardial disease. Its prognostic significance in coronary artery disease (CAD) remains uncertain.

OBJECTIVES The aim of this study was to assess the association between MRR and adverse cardiovascular outcomes across various CAD presentations.

METHODS A systematic review and meta-analysis was conducted according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. MEDLINE and Embase were searched from January 2019 to January 2025 for prospective studies reporting HRs for major adverse cardiovascular events in relation to MRR. Risk for bias was assessed using the Quality in Prognosis Studies tool. Pooled HRs were calculated using a random-effects model; heterogeneity was evaluated using the I^2 statistic.

RESULTS Five studies ($n = 3,186$) were included. Higher MRR was significantly associated with lower risk for adverse events (HR per unit increase: 0.75; 95% CI: 0.64-0.88; $I^2 = 80.9\%$). When dichotomized, low MRR conferred a more than 2-fold increased risk for major adverse cardiovascular events (HR: 2.39; 95% CI: 1.66-3.43). Subgroup analysis showed a stronger prognostic effect for ST-segment elevation myocardial infarction (HR: 0.46) vs stable CAD (HR: 0.86; P for interaction < 0.0001). Threshold analysis identified MRR ≥ 3 as optimal for sensitivity (58.9%) and rule-out performance, while lower thresholds improved specificity.

CONCLUSIONS MRR is a robust, independent predictor of cardiovascular outcomes of both acute and chronic CAD. Its prognostic impact is particularly pronounced for acute coronary syndrome. A threshold of 3 provides the best prognostic balance, supporting its integration into invasive physiological assessment for risk stratification.

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Coronary microvascular dysfunction (CMD) is a complex clinical entity that has been increasingly recognized as a major contributor to ischemic heart disease, particularly among patients with angina and nonobstructive coronary artery disease (ANOCA), as well as in other cardiovascular conditions influencing ischemic symptoms and prognosis.^{1–5}

Coronary flow reserve (CFR) has traditionally been used to assess coronary microvascular function. However, CFR is also influenced by epicardial coronary conductance, even in cases of nonobstructive disease (ie, diameter stenosis <50%), thus limiting its specificity for the microcirculation.^{6,7} To overcome this limitation, the index of microcirculatory resistance (IMR) was introduced as a measure of the minimal microvascular resistance, independent of the epicardial artery.⁸ IMR is measured during maximal hyperemia and does not take into account resting microvascular resistance.⁹ The microvascular resistance reserve (MRR) is a novel index designed to provide a more specific assessment of coronary microvascular function. It is defined as the ratio of true resting microvascular resistance, representing the hypothetical case of perfectly normal epicardial conductance, to measured hyperemic microvascular resistance.¹⁰ MRR is calculated according to the following formula:

$$\text{MRR} = \frac{\text{CFR} \cdot P_{a,\text{rest}}}{\text{FFR} \cdot P_{a,\text{hyper}}}$$

where FFR is fractional flow reserve and $P_{a,\text{rest}}$ and $P_{a,\text{hyper}}$ correspond to aortic pressure measured during resting and hyperemic conditions, respectively, thus correcting for any changes in driving pressure between nonhyperemic and hyperemic conditions. Given that the calculation of MRR adjusts for FFR, it is independent of epicardial resistance.¹¹ Yet, as CMD may coexist with epicardial coronary artery disease in both acute and chronic settings, MRR offers the potential to better discriminate between ischemia related to epicardial stenosis and that arising from microvascular dysfunction. However, the prognostic value of MRR remains incompletely understood, with only limited evidence available.

Accordingly, we conducted a systematic review and meta-analysis to evaluate the impact of abnormal MRR on major adverse cardiovascular events (MACE) and/or major adverse cardiovascular and cerebrovascular events (MACCE) in patients undergoing invasive coronary physiological assessment for suspected coronary artery disease.

METHODS

SEARCH STRATEGY AND INCLUSION AND EXCLUSION CRITERIA. We performed a systematic search of the MEDLINE and Embase databases from January 1, 2019, to January 30, 2025, for all studies in humans. The literature search strategy was designed by E.G. and P.Paolisso. Details of the search algorithm are provided in [Supplemental Table 1](#). This study was a meta-analysis of previously published data and did not involve new human subjects; therefore, Institutional Review Board approval and informed consent were not required.

We included studies that investigated the prognostic information of MRR and reported HRs for all-cause mortality and/or MACE and/or MACCE. There were no restrictions regarding language or sample size for the included studies.

Two investigators (E.G. and P.Paolisso) independently searched for eligible studies on the basis of the following inclusion criteria: 1) MRR assessed in at least 1 coronary artery; and 2) prognostic data available. Studies were excluded if measurements of clinical outcomes were not available or could not be extracted from the study. No restrictions were applied regarding the clinical presentation of the patients, with the inclusion of both acute coronary syndromes (ACS) and chronic coronary syndromes (CCS). When duplicates were encountered, the studies that had longer study periods were included and analyzed. Two authors performed the search and literature screening (F.D.L. and R.T.), with disputes resolved by discussion with a third author (E.G.). If studies had overlapping first or

ABBREVIATIONS AND ACRONYMS

ACS	= acute coronary syndrome(s)
ANOCA	= angina and nonobstructive coronary artery disease
CCS	= chronic coronary syndrome(s)
CFR	= coronary flow reserve
CMD	= coronary microvascular dysfunction
FFR	= fractional flow reserve
IMR	= index of microcirculatory resistance
MACCE	= major adverse cardiovascular and cerebrovascular event(s)
MACE	= major adverse cardiovascular event(s)
MRR	= microvascular resistance reserve
MVO	= microvascular obstruction
PCI	= percutaneous coronary intervention
STEMI	= ST-segment elevation myocardial infarction

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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senior authors, or institutions, E.G. and P.Paolisso evaluated the published methods to look for evidence of overlapping patient cohorts (dates, disease group, modality). Reference lists of included studies were screened to identify studies possibly missed from our searched results.

STATISTICAL ANALYSIS. The primary outcome measure was the HR for adverse clinical events per unit increase of MRR or comparing individuals with higher vs lower MRR values.

Two reviewers independently extracted data from each included study using a standardized form. Extracted variables included author, year of publication, study design, clinical presentation (categorized as ST-segment elevation myocardial infarction [STEMI] or CCS), total sample size, number of events, follow-up duration, and the reported HRs with corresponding 95% CIs. Discrepancies in data extraction were resolved by consensus.

Risk for bias was assessed at the study level using the Quality in Prognosis Studies tool, which is designed specifically for prognostic factor research. Each study was independently evaluated across 6 domains: study participation, study attrition, prognostic factor measurement (MRR), outcome measurement, confounding, and statistical analysis and reporting. Judgments were categorized as low, moderate, or high risk for bias for each domain. Disagreements between reviewers were resolved by consensus. Because of the observational design of all included studies, special attention was given to consistency in MRR measurement. Statistical analyses were conducted using the meta package (version 8.0.2) in R (R Foundation for Statistical Computing). Log-transformed HRs and corresponding SEs were calculated from the extracted effect estimates. A random-effects model based on the method of DerSimonian and Laird was used to pool effect sizes, accounting for between-study variability. Heterogeneity among studies was assessed using the Cochran Q test, the Higgins I^2 statistic, and the between-study variance τ^2 . I^2 values were stratified as <25%, 25% to 75%, and >75% for low, moderate, and high degrees of heterogeneity, respectively. To further explore potential sources of heterogeneity, we performed a subgroup analysis stratified by clinical presentation. Specifically, studies were categorized according to whether patients presented with STEMI or with CCS. Separate pooled HRs were calculated for each subgroup using a random-effects model, and a test for subgroup differences was applied to assess whether the prognostic value of MRR varied significantly between

clinical contexts. Sensitivity analyses were conducted by iteratively excluding individual studies and reassessing the pooled effect estimates to evaluate the robustness of the findings. Because of the limited number of included studies (<10), formal assessment of publication bias using funnel plots or statistical tests was not performed; nor were meta-regression analyses. All statistical tests were 2-tailed, with P values <0.05 considered to indicate statistical significance. The study was registered with the International Prospective Register of Systematic Reviews (CRD42025102228).

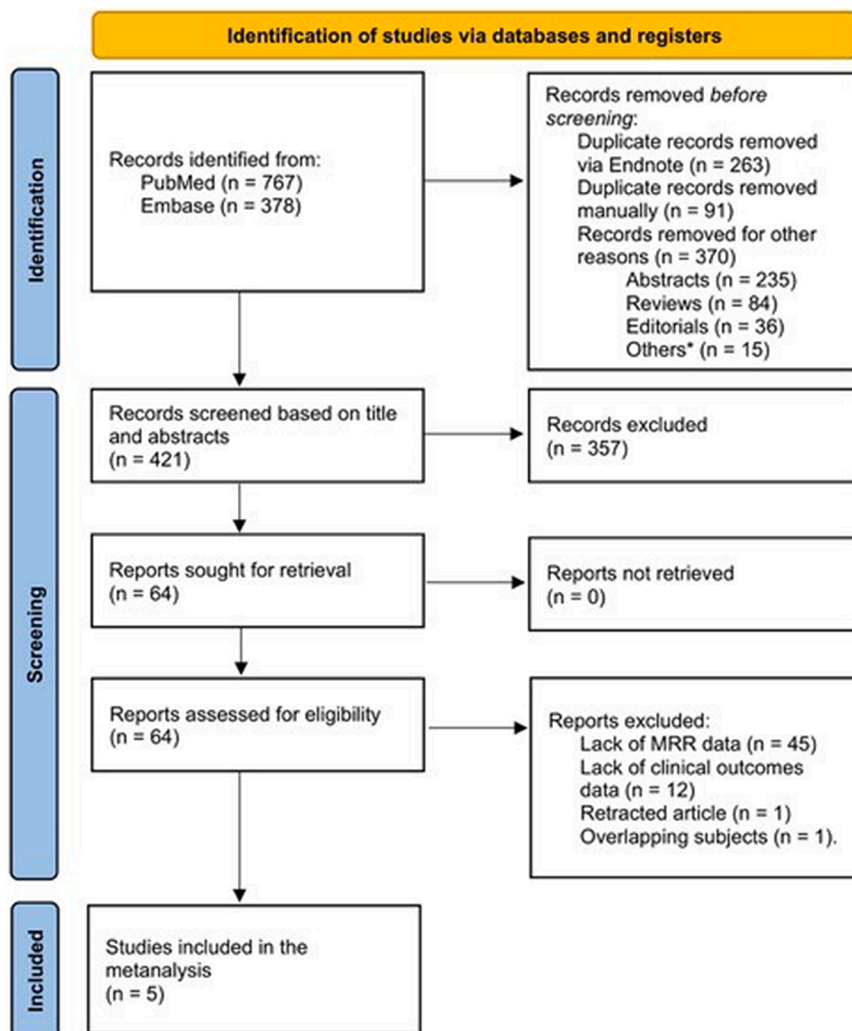
RESULTS

STUDY SELECTION AND POPULATION. Summaries of the study selection process and the search strategy are presented in [Figure 1](#) and [Supplemental Table 1](#), respectively. A total of 5 observational studies including 3,186 patients were ultimately included in the present meta-analysis.¹²⁻¹⁶ The main reasons for exclusion were the absence of MRR data and the absence of outcomes data related to the MRR. In 2 studies, MRR was measured in patients with STEMI following primary percutaneous coronary intervention (PCI) of the culprit vessel ($n = 656$); in another 2 studies, MRR was evaluated in patients with angiographically nonobstructive coronary artery disease or intermediate coronary stenoses ($n = 2,028$), and in 1 study, MRR was assessed after elective PCI ($n = 502$). MRR was calculated retrospectively in 3 studies^{12,14,15} and prospectively in 2 studies.^{13,16}

The pooled mean age was 63 ± 9.9 years and 75% of patients were men. MRR was assessed using bolus thermodilution except in 489 patients in whom intracoronary Doppler was used. The left anterior descending coronary artery was the most evaluated vessel (62.5%). Overall weighted mean MRR was 3.41 ± 1.34 , varying from 3.28 ± 1.53 in patients with CCS to 2.39 ± 0.76 in those with STEMI. Patient characteristics in the overall cohort and stratified by MRR values are reported in [Table 1](#) and [Supplemental Table 2](#). The weighted median follow-up across the 5 included studies was 3.37 years, with individual study medians ranging from 1.0 to 3.8 years. IQRs were reported by 4 of the 5 studies, varying from 1.4 to 6.1 years. Clinical outcomes definitions and median follow-up by each study are reported in [Supplemental Table 3](#).

The risk for bias across the 5 included studies was assessed using the Quality in Prognosis Studies tool. As shown in [Supplemental Figure 1](#), most studies showed a low risk for bias in nearly all domains. Boerhout et al¹² and Lee et al¹³ had the strongest

FIGURE 1 Study Selection



Preferred Reporting Items for Systematic Reviews and Meta-Analyses 2020 flow diagram outlining study selection for the meta-analysis. Of 1,145 records identified, 5 studies met inclusion criteria. *Other excluded sources included 3 clinical cases, 1 correction statement, 4 letters to the editor, 6 replies, and 1 research letter. MRR = microvascular resistance reserve.

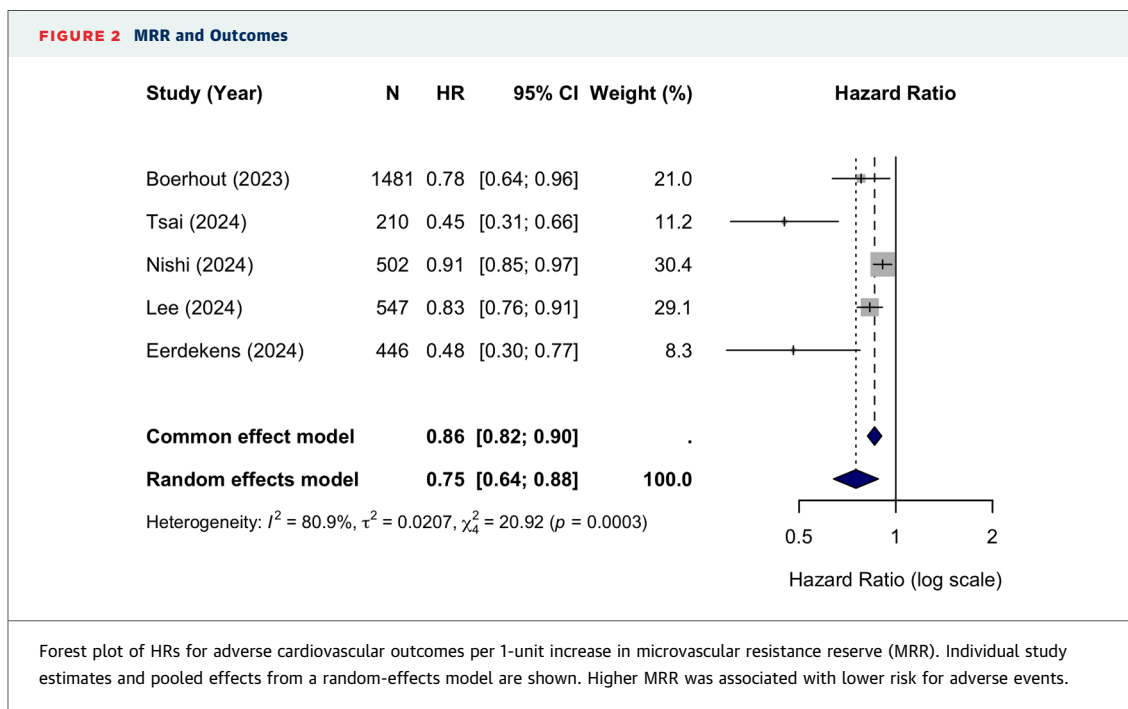
overall quality, with clear methods and good adjustment for confounders. A moderate risk for bias was identified in 2 studies: Eerdeken et al¹⁵ and Tsai et al¹⁶ had some concerns about study participation and confounding, and Tsai et al¹⁶ also had limitations in the statistical analysis due to the small sample size. Importantly, none of the studies was rated as high risk in any domain.

PROGNOSTIC VALUE OF MRR. When considering the MRR as a continuous variable to assess the dose-

TABLE 1 Baseline Characteristics in the Overall Population (N = 3,186)

Age, y	63 ± 9.9
Male	2,397 (75)
Diabetes	925 (29)
Hypertension	1,855 (58)
Dyslipidemia	1,926 (60.4)
Current smoking	1,022 (32)

Values are mean ± SD or n (%).



response relationship between MRR and adverse cardiovascular outcomes, HRs and corresponding SEs for each unit increase in MRR were extracted or calculated from eligible studies and log transformed prior to pooling. Using a random-effects model, the pooled HR per unit increase in MRR was 0.75 (95% CI: 0.64-0.88), indicating a 25% relative risk reduction for adverse events with each 1-unit increase in MRR. Substantial heterogeneity was detected across studies ($I^2 = 80.9\%$); the estimated between-study variance (τ^2) was 0.0207 (Figure 2, Supplemental Table 4).

Similarly, when considering the MRR as dichotomous variable according to the different cutoff values used in each study, low MRR was still consistently associated with an increased hazard of adverse cardiovascular outcomes. The pooled HR under a random-effects model was 2.39 (95% CI: 1.64-3.49) (Figure 3, Supplemental Table 5). There was considerable heterogeneity ($I^2 = 70\%$), reflecting variations in threshold definitions, populations, and methodologies.

SUBGROUP ANALYSIS. Subgroup analysis of studies measuring the association between MRR and clinical outcomes revealed a preserved, consistent effect across different clinical presentations (both STEMI and CCS) and low or moderate heterogeneity, though with markedly different effect sizes. Among CCS patients, the pooled HR for adverse outcomes per unit increase of MRR was 0.87 (95% CI: 0.80-0.94), with

moderate heterogeneity ($I^2 = 48.8\%$). In contrast, studies enrolling STEMI patients demonstrated a substantially stronger association (HR: 0.46; 95% CI: 0.34-0.62; $I^2 = 0\%$). The test for subgroup differences was statistically significant (chi-square = 15.64; $P < 0.0001$), with a more pronounced prognostic value of MRR in the setting of ACS (Figure 4).

Subgroup analysis of studies using different MRR thresholds showed that a threshold of $MRR \geq 3$ resulted in a pooled HR of 2.41 (95% CI: 1.64-3.55; $I^2 = 54.6\%$). In contrast, studies using a cutoff of $MRR < 3$ yielded a pooled HR of 2.45 (95% CI: 0.92-6.50), which did not reach statistical significance, though the direction of effect remained consistent ($I^2 = 88.2\%$). The test for subgroup differences was not statistically significant ($P = 0.981$) (Figure 5). Pooled diagnostic data from studies using a threshold of $MRR \geq 3$ ($n = 3$) and those using lower thresholds ($n = 2$) were descriptively compared. Studies using a threshold of ≥ 3 showed higher sensitivity (58.9%) but lower specificity (57.1%) compared with those using thresholds < 3 (sensitivity, 33.6%; specificity, 83.2%). The positive likelihood ratio was higher with thresholds lower than 3 (2.00 vs 1.37), while the negative likelihood ratio favored thresholds ≥ 3 (0.72 vs 0.80) (Table 2). These findings suggest that the threshold of 3 provides greater sensitivity and better rule-out performance, whereas lower thresholds offer improved specificity.

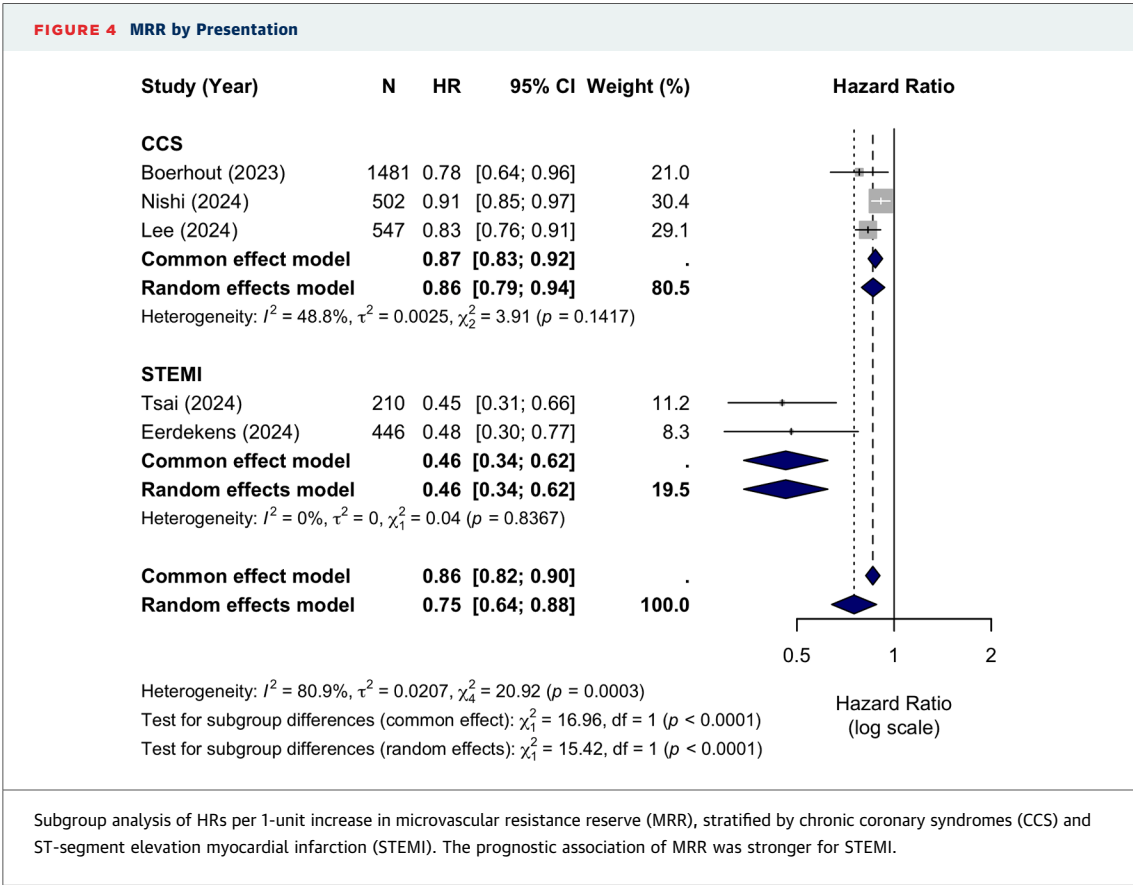
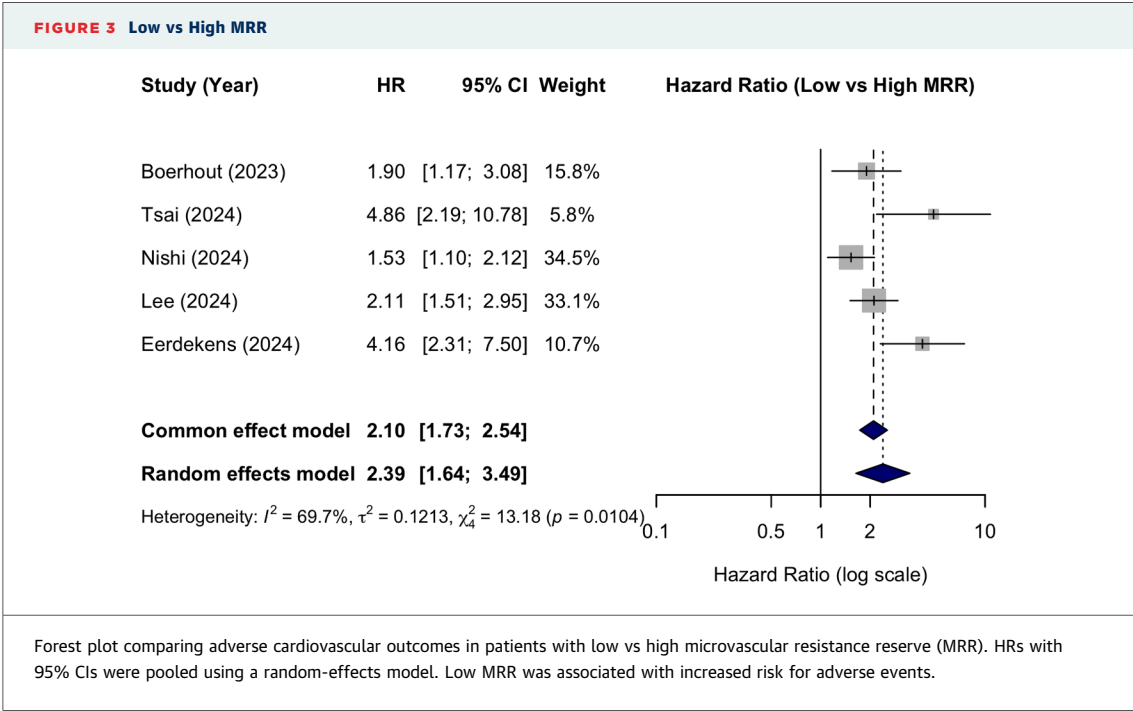
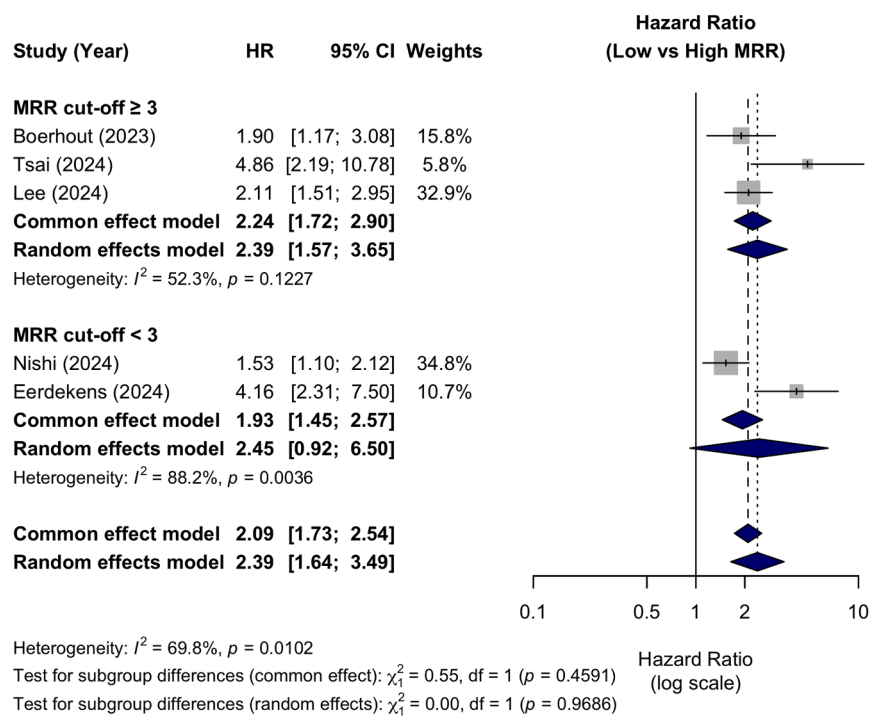


FIGURE 5 MRR Thresholds



Forest plot showing HRs for adverse cardiovascular outcomes according to different microvascular resistance reserve (MRR) cutoff values. An MRR threshold of 3 showed more consistent prognostic performance across studies.

SENSITIVITY ANALYSIS. To assess the robustness of our findings, we conducted leave-one-out sensitivity analyses using both continuous and dichotomous MRR. When modeled as a continuous variable, the association between higher MRR and improved clinical outcomes remained consistent across all iterations, with pooled HRs ranging from 0.66 to 0.82, and CIs that did not cross unity (Supplemental Figure 1). These results confirm that no single study disproportionately influenced the overall protective effect of higher MRR. Similarly, when MRR was analyzed as a dichotomous variable (low vs high), the pooled HRs remained significantly elevated across all leave-one-out iterations (range: 2.13-3.07), consistently demonstrating a higher risk for adverse events in patients with low MRR (Supplemental Figure 2). The exclusion of individual studies, including those with the most extreme HRs, did not materially alter the direction or significance of the pooled estimates in either analysis.

DISCUSSION

In the present study, we conducted a systematic review and meta-analysis of studies evaluating the association between MRR and clinical outcomes. The main findings were as follows: 1) MRR was associated with 25% risk reduction per each unit increase; 2) this effect was consistent for both patients with CCS and STEMI but significantly more pronounced in the latter setting; 3) considering the MRR as dichotomous variable, patients with low MRR had more than double risk for adverse outcomes compared with those with high MRR; and 4) although effect estimates were similar regardless of the MRR threshold used, a threshold of 3 provided a more consistent and precise effect size (Central Illustration).

METRICS FOR CMD. In clinical practice, nearly 40% of patients with angina are found to have ANOCA, a clinical condition that encompasses both vasospastic

TABLE 2 Diagnostic Performance of MRR Thresholds for Predicting Adverse Cardiovascular Outcomes

	MRR ≥3	MRR <3
Sensitivity (95% CI)	0.59 (0.54–0.64)	0.34 (0.28–0.40)
Specificity (95% CI)	0.57 (0.55–0.59)	0.83 (0.80–0.86)
Positive LR	1.37 (1.24–1.52)	2.00 (1.56–2.55)
Negative LR	0.72 (0.63–0.82)	0.80 (0.72–0.88)

Sensitivity, specificity, positive LR, and negative LR are reported with corresponding 95% CIs.
LR = likelihood ratio; MRR = microvascular resistance reserve.

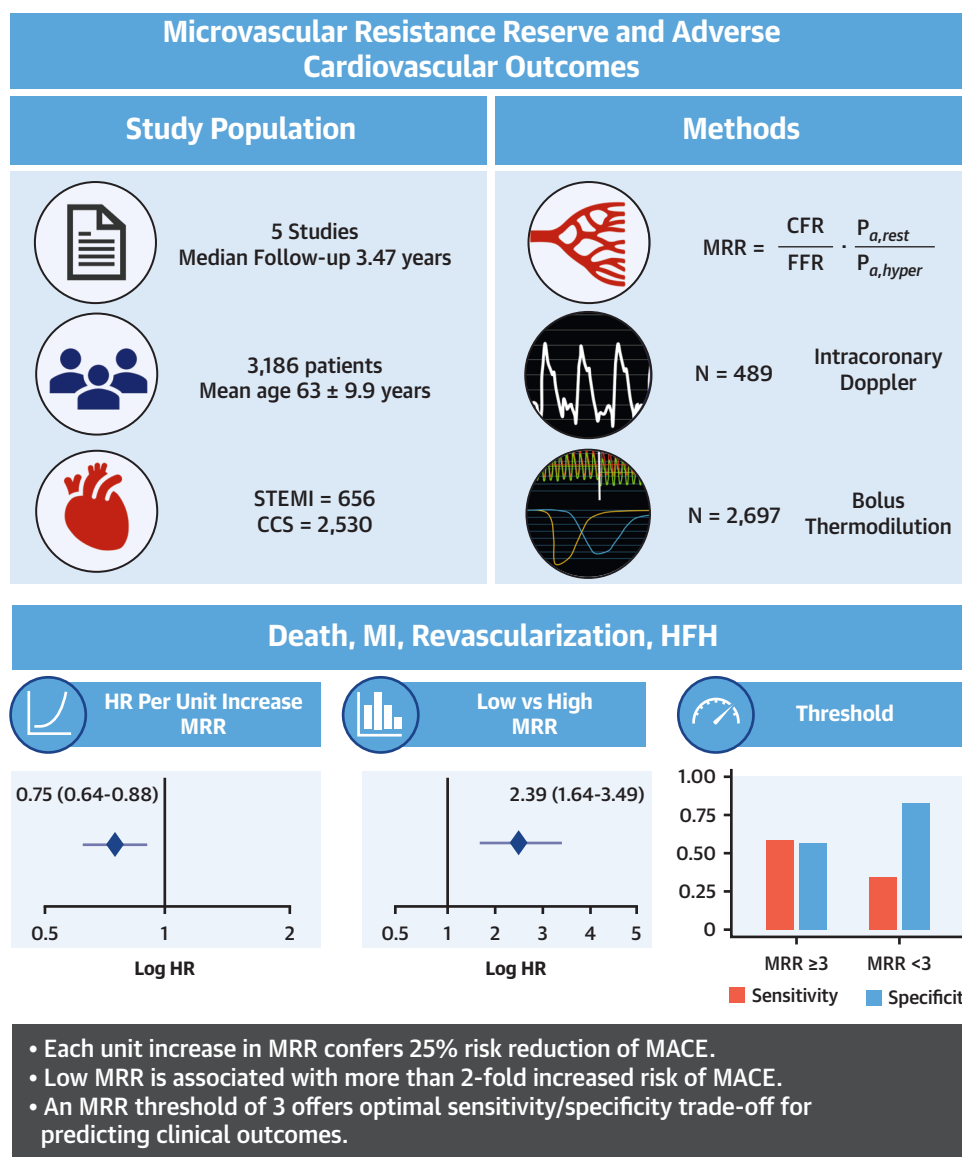
angina and CMD.¹⁷ The ideal metric in the identification of CMD should be highly sensitive, specific for coronary microcirculation and independent of epicardial stenoses, and both reproducible and validated. CFR, originally introduced to quantify the hemodynamic severity of epicardial coronary stenosis, has been used to define CMD under the assumption that impaired CFR in the presence of nonobstructive coronary artery disease is purely related to microvascular dysfunction.¹⁸ However, the assumption that angiographically nonobstructive vessels have negligible epicardial resistance is flawed, as even mild atherosclerosis can affect epicardial resistance, thereby affecting CFR and complicating the diagnosis of CMD.¹⁹ MRR, conversely, represents the ratio of true resting microvascular resistance and true hyperemic microvascular resistance being independent of epicardial stenosis; yet MRR is more specific for coronary microcirculation, allowing the assessment of CMD even in the presence of epicardial disease.^{11,20}

MRR AND CLINICAL OUTCOMES. The assessment of the prognostic value of MRR is still in the process of being developed. As seen from our systematic review, small studies have assessed the impact of abnormal MRR in different clinical settings. In the present pooled analysis, we found that each unit increase in MRR was associated with a nearly 25% reduction in risk for clinical events. Similarly, abnormal MRR conferred a 2.5-fold increased HR for clinical events. Overall, the prognostic value was consistent in patients with ACS and those with CCS, although it was significantly more pronounced in the former group. The heterogeneity across studies was high, likely reflecting differences in patient clinical presentations and risk profiles. Nevertheless, subgroup analysis based on clinical presentation confirmed a consistent association between MRR and clinical outcomes in the absence of statistical heterogeneity within each subgroup ($I^2 = 0\%$ for STEMI and $I^2 = 48.8\%$ for CCS), supporting the robustness of

these findings. Moreover, tests for subgroup differences confirmed a significantly greater prognostic effect of MRR in STEMI patients, with estimated 54% vs 14% risk reductions for STEMI and CCS, respectively ($P < 0.0001$). These findings suggest that microvascular (dys)function plays a more central role in determining outcomes following acute myocardial infarction.

In the setting of ACS, particularly STEMI, the occurrence and persistence of microvascular obstruction (MVO) after successful primary PCI, assessed using cardiac magnetic resonance, has been shown to correlate with suboptimal myocardial recovery, adverse left ventricular remodeling (predisposing to ventricular arrhythmias), as well as worse clinical outcomes.²¹ Yet IMR when assessed after primary PCI exhibits a weak correlation with MVO, with a discordance between IMR and MVO occurring in 36.4% of patients,²² although it has been shown to be a strong and independent predictor of cardiac death and all-cause mortality in this setting.^{23,24} However, whereas IMR is a measure of the minimal microvascular resistance, MRR investigates the vasodilatory capacity of the microcirculation. In this regard, Eerdekens et al¹⁵ showed that MRR had a good correlation with MVO, and a combination of both MRR and IMR outperformed either MRR alone or IMR alone in predicting clinical outcomes after primary PCI, and both were independent predictors of the composite of mortality or hospitalization for heart failure. Therefore, larger validation studies are needed to explore the accuracy of MRR and IMR as surrogate of MVO and intramyocardial hemorrhage, especially when cardiac magnetic resonance is not available, to improve risk stratification after primary PCI.

In the context of CCS, Boerhout et al¹² and Lee et al¹³ assessed the prognostic value of MRR in a cohort including patients with both obstructive and nonobstructive disease. Among patients with functionally significant epicardial stenosis, MRR, unlike CFR, was not significantly correlated with FFR, and particularly in patients with FFR <0.75, MRR was an independent predictor of MACE, whereas CFR was not.¹² Furthermore, among patients with intermediate epicardial stenosis, low MRR showed better predictability for the occurrence of MACE compared with CFR, IMR, or a combination of both.¹³ Last, the prognostic value of MRR was confirmed in the setting of elective PCI. Despite being a predictor of MACE, it was not associated with spontaneous myocardial infarction and target vessel revascularization; nevertheless, the association with all-cause death suggests that MRR might be indicative of overall health status beyond ischemic heart disease.¹⁴

CENTRAL ILLUSTRATION Meta-Analysis Evaluating the Prognostic Value of MRR

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CCS = chronic coronary syndromes; CFR = coronary flow reserve; FFR = fractional flow reserve; HFH = heart failure hospitalization; MACE = major adverse cardiovascular events (composite of all-cause [and/or cardiovascular] death, myocardial infarction, revascularization, and heart failure hospitalization); MI = myocardial infarction; STEMI = ST-segment elevation myocardial infarction.

However, this observation warrants further investigation in larger trials or registries.

CLINICAL IMPLICATIONS. The findings of this meta-analysis affirm the prognostic relevance of MRR, underscoring its potential utility in contemporary

cardiovascular risk stratification. We propose a threshold of 3 for use in clinical practice, as it holds greater sensitivity and better rule-out performance, whereas lower thresholds offer improved specificity at the cost of sensitivity, making them less reliable for early identification of at-risk patients. This is in

line with previous findings showing that an MRR of 2.7 virtually excludes the presence of CMD.²⁵ Therefore, the use of a standardized threshold of 3 may facilitate broader clinical adoption and improve risk stratification for patients with suspected CMD.

A metric that is both specific to coronary microcirculation and predictive of clinical outcomes holds significant clinical implications.

First, in the context of ACS, MRR may serve as a better metric for identifying patients at heightened risk for adverse events.

Second, CMD frequently coexists with epicardial coronary artery disease, and impaired microvascular vasodilation can exacerbate ischemic burden. Recent meta-analyses have demonstrated that CMD is independently associated with adverse cardiovascular outcomes, irrespective of the presence of obstructive epicardial lesions.²⁶ MRR, by offering a more precise and lesion-independent evaluation of microvascular health, may enhance prognostic stratification in patients with mixed coronary pathophysiology.

Third, post-PCI angina remains a significant clinical challenge, affecting 1 in 3 patients.²⁷ This residual symptom burden is associated with impaired quality of life, reduced physical function, anxiety, and increased health care use. Although delta FFR (defined as the difference between post- and pre-PCI FFR) has been shown to predict residual angina post-PCI, the mechanisms are likely multifactorial, with CMD implicated as a key contributor.^{27–30} Notably, the TARGET FFR (Post-stenting fractional flow reserve vs coronary angiography for optimization of percutaneous coronary intervention) trial reported that nearly one-third of patients experienced angina following PCI, despite no observed differences in baseline CFR.²⁷ Given that MRR and CFR represent distinct physiological constructs, with MRR reflecting a function of FFR, it is plausible that MRR could provide incremental diagnostic insight. An integrated approach combining FFR, the functional evaluation of the pattern of disease, as and the MRR may help identify those patients who may benefit from medical therapy targeting CMD. Such a strategy could be considered before proposing revascularization or at least to stratify the need for targeted antianginal therapy in addition to PCI.

STUDY LIMITATIONS. First, the number of included studies was relatively small, limiting statistical

power and generalizability. Although random-effects modeling was used, substantial heterogeneity across studies remains. Differences in patient populations, clinical settings, and MRR measurement techniques (bolus thermodilution vs Doppler) may have contributed to variability; additionally, the hyperemia protocols were not standardized. Notably, continuous intracoronary thermodilution has been recently introduced as novel method to quantify absolute coronary flow and microvascular resistance at rest and during hyperemia;^{31,32} although the method has been shown to provide higher reproducibility and accuracy than bolus thermodilution, none of the studies included in the present meta-analysis used this technique,^{20,33} thus limiting the applicability of our results to this methodology.

Second, thresholds for low MRR were inconsistent across studies, and follow-up durations varied, potentially affecting outcome comparisons. Specific components of MACE were not uniformly reported. Furthermore, quality-of-life outcomes (eg, angina relief, Seattle Angina Questionnaire scores, medication use) were not reported in the included studies. Future studies should incorporate standardized patient-reported endpoints to address this gap. Moreover, most included studies enrolled patients with established coronary artery disease, with limited data on those with ANOCA, limiting the generalizability of the findings to this growing patient population.

Third, meta-regression analysis and funnel plots to explore potential sources of heterogeneity could not be performed because of the small number of available studies.

Finally, although MRR is more specific for CMD than CFR, it remains an emerging index; prospective studies with standardized protocols are needed to validate its clinical utility; in this regard, further insights will be provided by ongoing Euro-CRAFT (European Microcirculatory Resistance and Absolute Flow Team) study (NCT05805462).

CONCLUSIONS

Impaired MRR is a strong predictor of adverse cardiovascular outcomes, with low MRR conferring a more than 2-fold increased risk for MACE or MACCE. An MRR threshold of 3 offers the optimal balance between sensitivity and specificity. These findings establish MRR as a robust metric for identifying CMD

and stratifying cardiovascular risk, independent of epicardial coronary disease.

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PERSPECTIVES

WHAT IS KNOWN? CMD is increasingly recognized as an important determinant of adverse cardiovascular outcomes; however, traditional indexes such as CFR are limited by confounding from epicardial disease, and the prognostic role of MRR remains incompletely defined.

WHAT IS NEW? This systematic review and meta-analysis provides the first quantitative synthesis of MRR's prognostic value, demonstrating that higher MRR is independently associated with a significant reduction in MACE, with greater predictive utility in ACS; an MRR threshold of 3 optimizes diagnostic performance.

WHAT IS NEXT? Prospective studies using standardized MRR acquisition methods are warranted to validate its role in risk stratification, particularly in the expanding population of patients with ANOCA, and to evaluate its integration alongside established physiological indexes in routine clinical practice.

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APPENDIX For extended methods, supplemental tables, figures, and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist, please see the online version of this paper.